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Lifecycle Cost Analysis and Business Case for High Integrity Protection System (HIPS) in Marjan Increment Program Onshore Oil Facilities

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Abstract

This comprehensive study presents a detailed lifecycle cost analysis and cost-benefit evaluation for implementing High Integrity Protection Systems (HIPS) in the Marjan Increment Program Package 6 (MIP-6) onshore oil facilities project. The analysis thoroughly compares the costs of HIPS with alternative solutions, such as facilities upgrades and additional capacity, to determine the most cost-effective approach. The results unequivocally show that HIPS offers significant cost savings over a 20-year operating lifetime, making it a highly attractive option. The study demonstrates a strong business case for the adoption of HIPS in the MIP-6 project, highlighting its potential to enhance safety and reduce costs. HIPS provides an additional layer of protection against hazardous events while resulting in substantial cost savings, thereby improving the overall efficiency of the project. The implementation of HIPS is a viable and cost-effective solution for the project's safety strategy, and its adoption is expected to yield long-term benefits.

Marjan Increment Program Overview

The Marjan Increment Program aims to keep Aramco as a reliable supplier for Oil and Gas, by increasing the Maximum Sustainable Capacity (MSC) target for the Marjan field from 500 MBCD to 800 MBCD of Arab Medium (AM) crude. In addition, it will provide facilities to process approximately 1,800 MMSCFD of incremental associated, non-associated, and cap gas from the Marjan field. Furthermore, for the first time, the Marjan Development Program designed and constructed a new Water Injection Plant (WIP) at the existing Tanajib Oil Plant to treat 650MBOD of water shipped from the Jaladi aquifer to be reinjected in the Marjan field to enhance the oil and gas production.

Given the fact that the Tanajib Oil Plant was constructed back in between 1983 and 1986 and commissioned by 1993, processing 500 MBCD of Arab Medium Crude, the Marjan Development Program came as the first major expansion for the Tanajib Oil Plant, increasing the footprint by 97% and the number of equipment by 76%, increasing crude processing capacity by 300MBCD, increasing gas production by 32 MMSCFD, as well as adding a new water supply and injection treatment plant to treat 650 MBOD by adding a new oil train, gas compression train, water treatment facility, utility facility, and five associated substations.

Background and Business Case

The Marjan Increment Program is a major onshore oil facilities expansion that requires the implementation of reliable and efficient safety systems to ensure the protection of people, the environment, and equipment. Within the project scope to expand the existing facilities and increase the production capacity, the age factor and limitations have been captured. And to meet the new production capacity while maintaining the latest safety and reliable industrial standards with effective engineering economic design, the project team conducted intensive engineering studies, such as a flare and relief study, HAZOP, SIL, and building risk assessment, that resulted in challenges founded in the existing facilities of the jump-over pipeline between existing facilities of 600# piping class and the new pipeline of 900# piping class, and overfilling risk scenarios at 2 low-pressure production traps and a flash drum. Where the risk has been captured and the mitigation found in either the upgrade of existing facilities through the replacement of equipment or the installation of additional safety and protection measures. The MIP-6 Project Team conducted a thorough lifecycle cost analysis and business case evaluation of upgrading existing facilities vs. the installation of High Integrity Protection Systems (HIPS).

LCC Methodology and Analysis Approach

The LCC analysis was conducted using a comprehensive model that considers procurement, operating, and maintenance costs, as well as the cost of failures. The model was based on the following steps:

- **Define the scope of the analysis:** The scope of the analysis was defined to include the costs associated with the implementation of HIPS, including procurement, installation, and maintenance costs.
- **Identify the alternatives:** The alternatives to HIPS were identified as facility upgrades and a replacement of the existing assets.
- **Estimate the costs:** The costs associated with each alternative were estimated, including procurement, installation, and maintenance costs.
- **Calculate the LCC:** The LCC was calculated using the following formula:

$LCC = C_P + C_O + C_F$; where

- C_P is the procurement cost.
- C_O is the operating cost.
- C_F is the cost of failures.

Procurement Cost (CAPEX)

The procurement cost, also known as the capital expenditure (CAPEX), represents the initial investment required to acquire and install the Safety Instrumented System (SIS). This cost is calculated using the following formula:

$$C_P = C_{HW} / L_{OP}$$

Where:

- C_P is the annual procurement cost (\$/yr).
- C_{HW} is the total hardware cost (\$), which includes the cost of engineering, equipment, installation, and commissioning.
- L_{OP} is the operating plant's lifetime (yrs.), assumed to be 20 years in this analysis.

The procurement cost calculation is based on the following assumptions:

- The SIS has a lifespan of 20 years, during which it will not experience any wear-out failures due to scheduled maintenance and good engineering practices.
- No discounting is applied to the procurement cost, as it is not considered relevant in this analysis.

By dividing the total hardware cost by the operating plant lifetime, we can determine the annual procurement cost, which represents the initial investment required to acquire and install the SIS.

Operating Cost (OPEX)

Operating costs are made up of scheduled and unscheduled maintenance costs. Scheduled maintenance includes proof testing, and unscheduled maintenance costs consist of the cost of repairing defects.

$$C_O = C_{SM} + C_{UM} + C_{ST}$$

Where:

- C_O is the annual operating cost (\$/yr).
- C_{SM} is the cost of scheduled maintenance (\$/yr), which includes the cost of regular maintenance activities, such as proof testing and calibration.
- C_{UM} is the cost of unscheduled maintenance (\$/yr), which includes the cost of repairing defects and failures.
- C_{ST} is the cost of spurious tripping (\$/yr), which includes the cost of unnecessary shutdowns and restarts.

The operating cost calculation is based on the following assumptions:

- Scheduled maintenance is performed regularly to prevent wear-out failures and ensure the SIS operates reliably.
- Unscheduled maintenance is required to repair defects and failures that occur during operation.
- Spurious tripping can occur due to various reasons, such as sensor faults or software glitches, and it can result in unnecessary shutdowns and restarts.

By adding up the costs of scheduled maintenance, unscheduled maintenance, and spurious tripping, we can determine the total annual operating cost, which represents the ongoing expenses associated with the operation and maintenance of the SIS.

Cost of Failure (C_F)

Given the fact that both options will eliminate the valuable consideration of the cost of failure, as the design of HIPS will consider redundancy and the option of fully rated replacement equipment will consider mitigation of failure. The cost of failure has been eliminated from the calculation.

- **Compare the alternatives:** The LCC of each alternative was compared to determine the most cost-effective solution.

The model was applied to three different scenarios:

- Overpressure HIPS vs. facility upgrade to fully rated piping
- Overfill HIPS vs. Replacement of New Separators Low-Pressure Production Traps (D-1 & D-2)
- Overfill HIPS vs. Replacement of new Flash Drum (D-14)

The analysis was based on input data and assumptions, including hardware costs, maintenance labour rates, and inventory revenue rates.

LCC Assumption

- Procurement costs are made up of the capital cost of the selected configuration of hardware. Estimated engineering costs have been included, and hardware costs are based on manufacturers' list prices.
- The model uses an operating lifetime of 20 years and assumes that wear-out failures do not occur within the SIS life. The assumption is that scheduled maintenance prevents wear-out, but no account is taken for the cost of item replacement within the SIS life since an allowance has been made for spares in procurement.
- The models assume no discounting with any interest rate.
- Scheduled maintenance consists of proof testing, and the cost associated depends upon the proof test frequency, test duration, maintenance labour rate, and the complexity of the system (its architecture), which will determine the number of proof tests to be carried out. Proof testing may also result in loss of revenue due to process shutdown.
- In turn, the cost of proof testing depends upon the number of tests required, the duration of the tests, and the labour rate for maintenance actions. The reliability model (HIPS SRS) will determine the number of proof tests required (NTP), based on the defined configuration and the proof test interval (TP).
- The cost of repairing defects that do not result in a process shutdown is determined by the defect rate, the mean downtime required for the repair, the replacement cost, and the maintenance labour rate.
- The cost of spurious shutdown is determined by the spurious trip rate, the mean downtime, and the inventory value.
- In both cases, the defect rate and spurious trip rate are determined by the system design and its architecture.
- An inventory value of zero has been used. It was assumed that production quotas would be made up from elsewhere in the field, and scheduled maintenance would take place during planned shutdown periods.
- Instrumented proof testing shall be 1/yr, and piping inspection shall be once every 5 years in line with the Layer of Protection Analysis (LOPA).

LCC Input Data

This study considered three models, each with two options: either to enhance safety by installing the HIPS or to upgrade the facilities.

- Model 1: Overpressure HIPS vs. facility upgrades to fully rated piping.
- Model 2: Overfill HIPS for LPPTs vs. New Separators.
- Model 3: Overfill HIPS for Flash Drum vs. New Flash Drum.

Overpressure HIPS vs. Facilities upgrade to fully rated piping

Table 1—Input Variables for Overpressure HIPS

Input Variable	Overpressure HIPS on Jump-over		
	Value	Units	Data Source
C_{HW} is the advised hardware cost	240,000	\$	Project EPC Records and Estimation
T_{TP} is the duration of proof testing	24	Hrs.	Project SIL studies
MTTR is the mean time to repair	168	Hrs.	Conservative estimation.
T_{STR} is the spurious trip rate	1.49E-02	/yr.	Project HIPS SRS
V_{INV} is the inventory revenue rate	1,000	\$/hr.	Industry Estimation
L_M is the maintenance labour rate	100	\$/hr.	Industry Estimation
N_{TP} is the number of proof test scheduled	1	/yr.	Project HIPS SRS
N_P is the number of staff required per test	4	People	Industry Estimation
TR_P is the number of trips (/yr)	1	/yr.	Project HIPS SRS
EX_T is the travel expense per trip (forward and back)	10,000	\$	Considered 5K USD per trip.
C_F is the cost of failure	0	\$/yr.	Not considered as Stated
T_{STR} is the spurious trip restoration time	168	/hr.	Project HIPS SRS
λ_{DEF} is the defect rate	1.00E-04	/yr.	Project HIPS SRS
L_{OP} is the operating plant lifetime	20	Yr.	Lifetime of 20 years assumed.

Table 2—Input variables for a fully rated piping system

Input Variable	Fully Rated Piping Option		
	Value	Units	Data Source
C_{HW} is the advised hardware cost	1.00E+07	\$	Project EPC Records and Estimation
T_{TP} is the duration of proof testing	168	Hrs.	Assumed to take the equivalent of one week for inspection of piping.
MTTR is the mean time to repair	168	Hrs.	Assumed to take the equivalent of one week for piping repair.
T_{STR} is the spurious trip rate	0.00E+00	/yr.	N/A.
V_{INV} is the inventory revenue rate	1,000	\$/hr.	Estimated.
L_M is the maintenance labor rate	100	\$/hr.	Considered 100\$ per hours based on industry Estimation
N_{TP} is the number of inspections scheduled	0.20	/yr.	Piping validation assumed every 5 years.
N_P is the number of staff required per test / maintenance activity	4	People	Industry estimation
TR_P is the number of trips (/yr)	0.2	/yr.	Piping validation assumed every 5 years.
EX_T is the travel expense per trip (forward and back)	0	\$	No additional expenses required as this would be part of normal T&I activities.
C_F is the cost of failure	0	\$/yr.	N/A since fully rated piping.
T_{STR} is the spurious trip restoration time	0	Hrs.	N/A.
λ_{DEF} is the defect rate as a result of unscheduled maintenance	1.00E-06	/yr.	As per Saudi Aramco SAEP-250 max. tolerable criteria has been applied.
L_{OP} is the operating plant lifetime	20	Yr.	Lifetime of 20 years assumed.

Overfill HIPS for LPPTs vs. New Separators (D-1 & D-2)

Table 3—Input Variables for Overfilling HIPS

Input Variable	Overfill HIPS at D-1 and D2		
	Value	Units	Data Source
C_{HW} is the advised hardware cost	336,000	\$	Project EPC Records and Estimation
T_{TP} is the duration of proof testing	24	Hrs.	Project SIL studies
MTTR is the mean time to repair	168	Hrs.	Conservative estimation.
T_{STR} is the spurious trip rate	1.49E-02	/yr.	Project HIPS SRS
V_{INV} is the inventory revenue rate	1,000	\$/hr.	Industry Estimation
L_M is the maintenance labour rate	100	\$/hr.	Industry Estimation
N_{TP} is the number of proof test scheduled	1	/yr.	Project HIPS SRS
N_P is the number of staff required per test	4	People	Industry Estimation
TR_P is the number of trips (/yr)	1	/yr.	Project HIPS SRS
EX_T is the travel expense per trip (forward and back)	10,000	\$	Considered 5K USD per trip.
C_F is the cost of failure	0	\$/yr.	Not considered as Stated
T_{STR} is the spurious trip restoration time	168	/hr.	Project HIPS SRS
λ_{DEF} is the defect rate	1.00E-04	/yr.	Project HIPS SRS
L_{OP} is the operating plant lifetime	20	Yr.	Lifetime of 20 years assumed.

Table 4—Input Variables for Fully Rated New Separator

Input Variable	New Separators at D-1 and D-2		
	Value	Units	Data Source
C_{HW} is the advised hardware cost	1.00E+07	\$	Project EPC Records and Estimation
T_{TP} is the duration of proof testing	168	Hrs.	LPPTs validation assumed every 5 years with duration of up to one week.
MTTR is the mean time to repair	168	Hrs.	Assumed to take the equivalent of one week for fixing / replacement of vessel with spares.
T_{STR} is the spurious trip rate	0.00E+00	/yr.	N/A.
V_{INV} is the inventory revenue rate	1,000	\$/hr.	Estimated.
L_M is the maintenance labour rate	100	\$/hr.	Considered 100\$ per hours based on industry Estimation
N_{TP} is the number of inspections scheduled	0.20	/yr.	LPPT validation assumed every 5 years.
N_P is the number of staff required per test / maintenance activity	4	People	Industry estimates.
TR_P is the number of trips (/yr)	0.2	/yr.	LPPT validation assumed every 5 years.
EX_T is the travel expense per trip (forward and back)	0	\$	No additional travel costs expected as the vessel can be maintained along with the existing vessels as part of normal T&I.
C_F is the cost of failure	0	\$/yr.	N/A since the additional vessel will increase capacity to maintain full inventory for a significant period.
T_{STR} is the spurious trip restoration time	0	/hr.	N/A.
λ_{DEF} is the defect rate as a result of unscheduled maintenance	1.00E-06	/yr.	SAEP-250 [7] max. tolerable criteria have been applied.
L_{OP} is the operating plant lifetime	20	Yr.	Lifetime of 20 years assumed.

Overfill HIPS for Flash Drum vs. New Flash Drum

Table 5—Input Variables for Overfilling HIPS

Input Variable	Overfill HIPS at Flash Drum (D-14)		
	Value	Units	Data Source
C_{HW} is the advised hardware cost	650,000	\$	Project EPC Records and Estimation
T_{TP} is the duration of proof testing	24	Hrs.	Project SIL studies
MTTR is the mean time to repair	168	Hrs.	Conservative estimation.
T_{STR} is the spurious trip rate	1.35E-02	/yr.	Project HIPS SRS
V_{INV} is the inventory revenue rate	1,000	\$/hr.	Industry Estimation
L_M is the maintenance labor rate	100	\$/hr.	Industry Estimation
N_{TP} is the number of proof test scheduled	1	/yr.	Project HIPS SRS
N_P is the number of staff required per test	4	People	Industry Estimation
TR_P is the number of trips (/yr)	1	/yr.	Project HIPS SRS
EX_T is the travel expense per trip (forward and back)	10,000	\$	Considered 5K USD per trip.
C_F is the cost of failure	0	\$/yr.	Not considered as Stated
T_{STR} is the spurious trip restoration time	168	/hr.	Project HIPS SRS
λ_{DEF} is the defect rate	1.00E-04	/yr.	Project HIPS SRS
L_{OP} is the operating plant lifetime	20	Yr.	Lifetime of 20 years assumed.

Table 6—Input Variables for Fully Rated New Flash Drum

Input Variable	New Flash Drum		
	Value	Units	Data Source
C_{HW} is the advised hardware cost	1.50E+7	\$	Email from HDEC.
T_{TP} is the duration of proof testing	168	Hrs.	Vessel validation assumed every 5 years with duration of up to one week.
MTTR is the mean time to repair	168	Hrs.	Assumed to take the equivalent of one week for fixing / replacement of vessel with spares.
T_{STR} is the spurious trip rate	0.00E+00	/yr.	N/A.
V_{INV} is the inventory revenue rate	0	\$/hr.	N/A since considered to be always available.
L_M is the maintenance labour rate	100	\$/hr.	Estimated.
N_{TP} is the number of inspections scheduled	0.20	/yr.	Vessel validation assumed every 5 years.
N_P is the number of staff required per test	4	People	Industry estimates.
TR_P is the number of trips (/yr)	0.2	/yr.	Vessel validation assumed every 5 years.
EX_T is the travel expense per trip (forward and back)	0	\$	No additional travel costs expected as the vessel can be maintained along with the existing vessels as part of normal T&I.
C_F is the cost of failure	0	\$/yr.	N/A since the additional vessel will increase capacity to maintain full inventory for a significant period.
T_{STR} is the spurious trip restoration time	0	/hr.	N/A.
λ_{DEF} is the defect rate as a result of unscheduled maintenance	1.00E-06	/yr.	As per Saudi Aramco SAEP-250 max. tolerable criteria has been applied.
L_{OP} is the operating plant lifetime	20	Yr.	Lifetime of 20 years assumed.

LCC Analysis and Calculation

- Based on the results of the LCC analysis for Model 1, there are significant savings in utilizing HIPS as a means of overpressure protection in comparison to upgrading the facilities to fully rated piping. The Cost Benefit Analysis (CBA) suggests savings of up to 81% (\$58,355/yr. vs. \$547,040/yr.), and this presents a solid business case.
- Based on the results of the LCC analysis for Model 2, there are notable savings in utilizing the overfill HIPS as a means of overfill protection in the evaluation of additional LPPTs for D-1 and D-2, and thus the BC suggests savings of up to 88% (\$63,155/yr. vs. \$547,040/yr.). Similarly, the BC economically favors the overfill HIPS.
- Based on the results of the LCC analysis for Model 3, there are considerable savings in utilizing the overfill HIPS as a means of overfill protection in the evaluation of an additional vessel for D-14. The BC suggests savings of up to 90% (\$78,596/yr. vs. \$797,040/yr.). Similar to models 1 and 2, the BC provides a strong suggestion for the overfill HIPS.

Table 7—LCC Analysis Calculation

	Model 1		Model 2		Model 3	
	Overpressure HIPS	Facilities Upgrade	Overfill HIPS	Capacity Increase	Overfill HIPS	Capacity Increase
Total Annual Cost (\$/yr)	58,355	547,040	63,155	547,040	78,596	797,040
$LCC = C_P + C_O + C_F$	58,355	547,040	63,155	547,040	78,596	797,040
C_P is the cost of plant (\$/yr);	12,000	500,000	16,800	500,000	32,500	750,000
C_O is the cost of operation (\$/yr);	46,355	47,040	46,355	47,040	46,096	47,040
C_F is the cost of failures (\$/yr).	-	-	-	-	-	-
Cost Saving with HIPS Option	81%		88%		90%	

Conclusion

Lifecycle cost analysis confirms that High Integrity Protection Systems (HIPS) offer significant cost savings compared to traditional facility upgrades. Across all three models, HIPS implementation resulted in savings ranging from 81% to 90%, demonstrating a strong business case for their adoption. The analysis was based on a 20-year operating life, with scheduled maintenance preventing wear-out failures. No discounting was applied, and the cost of failure was excluded due to the high reliability of both HIPS and upgraded systems. From a business perspective, HIPS provide a compelling alternative to facility upgrades, offering equivalent safety and reliability at a fraction of the cost.

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